

1. PORTAL HOMEPAGE & OVERVIEW ARCHITECTURE

Central entry hub for MSc Integrated Mathematics, Data Science & Computing students and faculty.



CIT Cognitive Assessment Portal - Desktop Container

MSc Integrated Mathematics Cognitive Assessment System

An advanced 50-question 60-minute standardized evaluation across 5 core mathematical domains.

[+] 50 Questions / 5 Sections

Numerical, Verbal, Algorithmic, Working Memory, Abstract

[+] 60 Minutes Duration

Strict countdown with automatic session submission

[+] 40% - 30% - 30% Difficulty

4 Easy, 3 Moderate, 3 Hard questions per section

[+] Real-Time Auto-Save

Responses synced to browser storage every 30 seconds

Core Assessment Domains & Question Distribution:

1. Numerical Reasoning & Applied Calculus

Calculus, matrices, linear algebra, sequence convergence, probability distributions.

2. Verbal & Logical Inference

Data interpretation, logical deduction, syllogisms, mathematical statement validation.

3. Algorithmic & Computational Thinking

Graph theory, recurrences, time complexity, discrete structures, optimization algorithms.

4. Working Memory & Reflective Response

Pattern retention, multi-step problem solving, error detection, mental calculation speed.

5. Abstract & Spatial Reasoning

Geometric transformations, spatial rotations, topological properties, matrix patterns.

2. STUDENT LOGIN & CANDIDATE REGISTRATION PORTAL

Form layout, validation rules, input fields, and pre-examination instructions modal.

Student Onboarding Form Layout (UI Wireframe Representation):

STUDENT FULL NAME *

Input Example: "Vaidehi S" [Required Text Field]

REGISTER NUMBER / ROLL NO *

Input Example: "22MSR045" [Auto-Capitalized on Input]

ACADEMIC DEPARTMENT *

Dropdown Select: [MSc Data Science | MSc Software Systems | MSc Decision & Computing Sciences]

STUDENT ACCESS PASSCODE *

Default Access Key: "CIT-2026" [Validation Guard]

START CANDIDATE ASSESSMENT (60 MINS)

Candidate Test Guidelines & Pre-Examination Rules:

1. Test Format & Structure: Exactly 50 multiple-choice questions divided into 5 equal sections.
2. Timer & Countdown: 60 minutes countdown begins immediately upon clicking start button.
3. Synchronized Launch: Includes a 10-second candidate readiness countdown overlay before initial Q1 load.
4. Section Navigation: Free toggle between all 5 sections at any point during test duration.
5. Question Status Flagging: Questions can be flagged for review and tracked in the status palette.
6. Auto-Save Security: Candidate response state saved every 30 seconds to prevent data loss.
7. Time Extension Request: Candidates may request a +5 minute extension subject to admin rule settings.
8. Immediate Analytics: Upon submission, candidate receives detailed cognitive radar & section metrics.

3. FACULTY LOGIN & MASTER ADMIN PORTAL

Faculty authentication layout, PIN security guards, question management & evaluation control.

Faculty Authentication Form Layout (UI Wireframe Representation):

FACULTY INSTITUTIONAL EMAIL *

Input Example: "faculty@cit.edu.in" [Institutional Domain Filter]

MASTER ADMIN SECURITY PIN *

Default Security PIN: "CITADMIN2026" [High Security Access Gate]

AUTHENTICATE & ENTER FACULTY PORTAL

Faculty Administrator Features & Controls Summary:

1. Question Bank Management

Create, edit, delete, and categorize questions across all 5 cognitive domains with difficulty tags.

2. Live Assessment Configurator

Adjust time limits (default 60 mins), section weights, difficulty ratios, and pass thresholds.

3. Master Assessment Report Generator

Generate datewise, departmentwise, and candidatewise summary analytics and CSV/Excel exports.

4. Candidate Performance Analytics

View colorful domain distribution charts, accuracy rates, average response time, and cognitive levels.

5. AI Cognitive Synthesis Engine

Generates automated Gemini AI qualitative candidate evaluations and research archetype recommendations.

6. Institutional Security & Controls

Access tier controls, password resets, activity logs, and secure data backup/restore utilities.